

“Emphasis is on carbon emission reduction, safety and connected gadgets inside automobiles”



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Delphi is a leading global supplier of electronics and technologies for automotive, commercial vehicle and other segments. It delivers innovations that make products smarter and safer as well as more powerful and efficient.

Delphi's Technical Center India (TCI), established in Bengaluru in 2000, is one of the largest Delphi technical centres outside USA. Tarak Balaji, director, Technical Center India, Delphi Automotive Systems, spoke to EFY's Jesus Milton Rousseau S. about the contributions of this technical centre and their future plans.

Please tell us more about Bengaluru technical centre.

Delphi India's TCI complements Delphi's strong manufacturing base in India and in the rest of the Asia Pacific region. Advanced engineering activities at Delphi India's TCI include embedded software, mechanical engineering, product engineering, and design and engineering. Delphi India is steadily increasing product engineering capabilities to keep pace with the increasing use of electronics in vehicles.

What are the latest trends in automobile designing?

There is a significant growth of 8 to 20 per cent in electric cars and increased penetration of electronics in automobiles. Also, small cars are starting to use electronics.

In the automotive industry, there is emphasis on reducing the carbon emission by using sophisticated engine electricals and electronics with the new standards coming up, automotive safety and use of connected gadgets like navigation, MP3 players and other entertainment devices inside the vehicles.

Can you name products customised for Indian market?

We have products like infotainment modules (Internet radio, TV), vehicle control modules (body control modules) and anti-theft modules ready for the Indian market.

What is your digital drive assist system (DDAS) for Mahindra & Mahindra's multipurpose vehicle XYLO all about?

The advanced DDAS is an alphanumeric display centrally mounted on the vehicle's dashboard having a set of unique features that compute important information for the driver instantaneously, while on the move. The system calculates such information as mileage and distance permissible with available fuel. Standard information such as date, time, inside/outside temperature, relative humidity, fuel economy, vehicle speed and gear position are available at a

quick glance. Further, a 3-D digital compass lets the driver know the exact direction in which the car is headed. The system also displays average speed, instant mileage clocked and fuel-efficiency, if desired. The inbuilt warning system provides alerts to the driver for seat belts, handbrake on, vehicle over-speed, door ajar, low brake oil, high engine temperature, low engine oil pressure, water in fuel, battery not charging, air-filter clogged, clock alarm, malfunction-indicator lamp-on and malfunction-indicator lamp-blink.

How has been the experience of working for Tata Nano?

Delphi is supplying instrument cluster and immobilisers for Tata Nano. Joint activities with Tata's engineering and commercial teams started at the concept stage and continued till production-intent designing stage. The joint team developed electrical and mechanical interfaces for the vehicle, scripting and formulation of critical algorithms, and many other technical details.

Cost reduction ideas started emerging from vehicle interface definitions to material selection for plastic enclosures and printed circuit boards. The instrument cluster was kept basic with just a speedometer, odometer and turn-indicator signals. Fuel gauge was integrated with display and the temperature indication was done through LED indicator instead of the conventional gauge.

What are the projects at R&D stage?

We are working on next-generation technology products ranging from alternative fuel-powered systems to instrument cluster with colour TFT screen to touchscreen-based navigation.

What is the percentage of your revenue from exports?

We have 80 per cent revenue from exports and only 20 per cent from domestic customers.